# Miguel Zumalacárregui

Ramón y Cajal Fellow — Instituto de Física Teórica IFT-UAM/CSIC, Madrid
Group Leader, ACR Department — Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam

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The publication record is maintained on INSPIRE-HEP; research projects are described on the website.

## Positions

- **Ramón y Cajal Fellow**
  Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2026–
- **Group Leader**, Astrophysical and Cosmological Relativity
  Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam · 2020–
- **Marie Skłodowska-Curie Global Fellow**
  Berkeley Center for Cosmological Physics, with return phase at IPhT Saclay · 2017–2020
- **Nordita Fellow**
  Nordic Institute for Theoretical Physics, Stockholm · 2015–2017
- **Postdoctoral Researcher**
  Institut für Theoretische Physik, University of Heidelberg · 2013–2015
- **Postdoctoral Researcher**
  Instituto de Física Teórica IFT-UAM/CSIC, Madrid · 2013

## Education

- **PhD in Physics**, *Cum Laude* with International Mention
  Universidad Autónoma de Madrid · 2012 · Graduate work at ICC Barcelona and IFT-UAM/CSIC (FPI fellowship)
  Dissertation: Probing the Foundations of the Standard Cosmological Model · defense presentation
- **Master in Theoretical Physics**
  Universidad Autónoma de Madrid · 2009
  Thesis: Disformal Quintessence: Observational Constraints and Linear Perturbations

## Supervision

### Postdoctoral researchers

- **Daniel Ballard** — from November 2026
- **Han-Gil Choi** — 2026–
- **Xikai Shan** — Affiliated postdoctoral researcher, 2026–
- **Juan Urrutia** — from November 2026
- **Héctor Villarrubia-Rojo** — 2026–
- **Srashti Goyal** — 2023–2026 → Independent Fellow at IUCAA
- **Serena Giardino** — 2024–2025 → Postdoc at Imperial College London
- **Guilherme Brando** — Humboldt Fellow, 2022–2024 → Tenured Researcher at UFES, Brazil
- **Giovanni Tambalo** — 2020–2023 → Postdoc at ETH Zürich, then Marie Skłodowska-Curie Fellow at IPhT Saclay

### PhD students

- **Emanuela Alberici** — from October 2026

- **Manon van Zyl** — from October 2026
- **Nicola Menadeo** — PhD student, U. Naples (co-supervised), 2023–2025, long-term stay → Postdoc at Humboldt University of Berlin
- **Stefano Savastano** — PhD student (co-supervised with A. Buonanno), 2020–2024 → PhD awarded at HU Berlin
- **Carlos García-García** — PhD student, IFF-CSIC (co-supervised), 2017–2020 → Beecroft Fellow at Oxford, now tenure-track scientist at CIEMAT

### Master's students

**Matteo Pagani** (U. Bologna, co-supervised, 2026); **Louis Salimbeni** (ETH Zürich, co-supervised, 2026); **Pablo Vega del Castillo** (LMU, 2026); **Shashwat Singh** (Paris Observatory, 2022–2023 → PhD at the University of Glasgow); **Julius Streibert** (FU Berlin, co-supervised, 2022–2023); **Richard Stiskalek** (LMU, co-supervised, 2021–2022 → PhD at Oxford, pre-doctoral Fellow at CCA, now postdoc at Oxford); **Alessandro Longo** (U. Trieste, co-supervised, 2020–2021 → PhD student at SISSA); **Janina Renk** (U. Heidelberg, co-supervised, 2015–2016 → PhD at Stockholm University).

### Research interns and undergraduates

**Selin Üstündag** (2026 → PhD at Newcastle University); **Marcel Glätzer** (HU Berlin, 2024–2025); **Jaime Raynaud** (TU Berlin & KTH, 2024–2025 → PhD at MPIA); **Vaibhav Omble** (LMU, 2023–2024); **Daryna Yushchenko** (Princeton summer program, 2024 → PhD at Carnegie Mellon University); **Lyla Choi** (Princeton summer program, 2023 → PhD at Stanford); **Yukei Murakami** (UC Berkeley, 2018–2019 → PhD at Johns Hopkins University); **Brindha Kanniah** (MIT summer program in Heidelberg, 2015 → Master's at MIT).

Visiting researchers, and students mentored informally rather than formally supervised, are listed at miguelzu-malacarregui.es/group.html.

## Scientific coordination

- **LISA Consortium**
  member since 2018
  Council member and Senior Co-Chair of the Cosmology Working Group (2025–); section coordinator for gravitational-wave lensing in the LISA Cosmology white paper (2022)
- **Euclid Consortium**
  member since 2014
  Co-lead of the Science Working Group on Gravitational Waves and of its "bright sirens" work package (2025–); member of the Strong Lensing, Cosmology Theory, and Supernovae & Transients working groups
- **GW-Space 2050**
  Coordinator for the cosmography section, defining ESA's gravitational-wave missions after LISA (2023–)
- **Multi-messenger lensing white paper**
  Coordinator for the microlensing section (2024)

## Workshops and meetings organized

- 13th LISA Cosmology Working Group Workshop
  Barcelona, 2026
- 12th LISA Cosmology Working Group Workshop
  Tallinn, 2025
- Lensing and Wave Optics in Strong Gravity
  Erwin Schrödinger Institute, Vienna, 2024

- LISA joint Fundamental Physics + Waveform meeting
  Potsdam, 2024
- ACR Division Seminars
  Max Planck Institute for Gravitational Physics, 2020–2022
- BCCP Cosmology Lunch
  UC Berkeley, 2016–2017
- Cosmology Seminars, Institut für Theoretische Physik
  University of Heidelberg, 2014–2015
- TRR33 Winter Schools
  Passo del Tonale, 2013, 2014 and 2015

## Teaching

### Graduate schools

- **Gravitational waves**, TAE (Taller de Altas Energías), Benasque, 2024
  Slides
- **Tests of gravity and dark energy with cosmology and gravitational waves**
  II Mesoamerican Workshop on Gravity and Cosmology, Chiapas, 2018
- **Alternative gravity theories**, Cosmology School in the Canary Islands, 2017
  Video lecture
- **School for Cosmology Tools**, IFT Madrid, 2017
  Lectures on CLASS, hi_class and MontePython; material on the Talks & Teaching page
- **Workshop on CLASS and MontePython**, MPIA Munich, 2014

### University courses

- **Berkeley Connect Instructor**, UC Berkeley, 2018–2019
  Two semesters of undergraduate mentoring and teaching
- **Lecturer**, University of Heidelberg, 2014–2015
  Advanced Cosmology; Introduction to CMB Physics; Statistical Mechanics; advanced seminar
- **Teaching assistant**, University of Heidelberg, 2014–2015
  Theoretical Astrophysics (graduate), with M. Bartelmann
- **Teaching assistant**, Universidad Autónoma de Madrid, 2013
  Gravitation and Cosmology; Physics of the Cosmos
- **Mini-course on inhomogeneous cosmologies**, University of Barcelona, 2012

## Funding

- **ERC Consolidator Grant**, GLOW: Gravitational Lensing of Waves · 2,185,007 €
  Principal Investigator · European Research Council · 2026–2031
  Project site: glow-astro.org
- **Ramón y Cajal Fellowship**, ranked 2nd in Physics · IFT-UAM/CSIC · 2026–
  Atracción de Talento award · Agencia Estatal de Investigación · 100,000 €
  RyC-MaX excellence funding · CSIC · 225,000 €
- **Max Planck Society Annual Project**, With Einstein on Crooked Paths · 286,940 €
  Principal Investigator · funded by philanthropic donations · 2024–
- **Cosmology group core funding**, Max Planck Institute for Gravitational Physics
  Group Leader · Astrophysical and Cosmological Relativity division · 2020–
  Two postdocs and one PhD student

- **Workshop funding**, Lensing and Wave Optics in Strong Gravity · 12,000 €
  Erwin Schrödinger Institute, Vienna · 2024
- **Marie Skłodowska-Curie Global Fellowship**, NLO-CO · 264,668 €
  European Commission · UC Berkeley and IPhT Saclay · 2015–2020
- **FPI doctoral fellowship** · 54,816 € and 14,550 € for research stays
  Spanish Ministry of Science · stays in Heidelberg and Oslo · 2008–2012
- **Host** for two awarded Humboldt Fellowships (2022, 2025) and one Margarita Salas Fellowship (2022)

## Other awards and fellowships

- R3 certification, Spanish Research Agency.
- Berkeley Connect Fellow, UC Berkeley (2018).
- Nordita Fellowship (2015).
- Yggdrasil Fellowship (2011): research stay at ITA Oslo.
- UAM Master's scholarship (2007).

## Evaluation and peer review

### Evaluation panels

- NASA Hubble Fellowship Program, Physics and Cosmology panel.
- FCT (Portuguese Foundation for Science and Technology), Individual Support Call.
- Max Planck Institute for Gravitational Physics, postdoctoral hiring committee, since 2020.

### External evaluator for funding agencies

European Research Council (Starting Grants), STFC/UKRI, Swiss National Science Foundation, Alexander von Humboldt Foundation, InterTalentum (Marie Skłodowska-Curie COFUND), Netherlands Research Council (NWO), Research Grants Council of Hong Kong, and the Estonian Research Council.

### Refereeing

Peer review for more than 40 papers, including Physical Review Letters, Physical Review D, Nature Astronomy, JCAP, The Astrophysical Journal and Astrophysical Journal Letters.

## Selected talks

- **Multi-Messenger Lensing Program**, Lorentz Center, Leiden, 2026 · plenary
  Don't kill the messenger! New phenomena, new challenges, new opportunities
- **Universidad Complutense de Madrid**, IPARCOS, 2026 · institute colloquium
  Gravitational lensing of waves
- **IAS and Princeton University**, gravitational-wave meeting, Princeton, 2026 · seminar
  GW231123 as a diffracted and magnified black-hole merger
- **Leibniz Institute for Astrophysics**, Potsdam, 2024 · institute colloquium
  Gravitational lensing of waves
- **New Frontiers in Strong Gravity**, Benasque, 2024 · plenary
  Wave-optical phenomena in gravitational-wave lensing
- **VIII Meeting on Fundamental Cosmology**, Granada, 2022 · plenary
  Gravitational-wave lensing and fundamental physics
- **EuCAPT Annual Symposium**, CERN, 2022 · plenary review
  Dark energy and tests of gravity with cosmology and gravitational-wave propagation

- **Progress on Old and New Themes in Cosmology**, Avignon (online), 2020 · plenary
  Tests of gravity with gravitational waves
- **COSMO**, Aachen University, 2019 · plenary
  Tests of dark energy and modified gravity
- **Rencontres du Vietnam**, Very High Energy Phenomena in the Universe, 2018 · plenary
  The dark Universe in the era of gravitational-wave astronomy

## Languages

Spanish (native), English (C2), German (C1).

## Selected publications

The complete record is on INSPIRE-HEP.

- Effective description of lensed gravitational waves diffracted by stellar fields
  M. Zumalacárregui, X. Shan · Phys. Rev. D 114, 063038 (2026) · Editors' Suggestion
- Across the Universe: GW231123 as a magnified and diffracted black hole merger
  S. Goyal, H. Villarrubia-Rojo, M. Zumalacárregui · Astrophys. J. Lett. 1008, L12 (2026)
- Black holes as telescopes: discovering supermassive binaries through quasi-periodic lensed starlight
  H. Wang, M. Zumalacárregui, B. Kocsis · Phys. Rev. Lett. 136, 061403 (2026) · Editors' Suggestion
- Gravitational wave lensing as a probe of halo properties and dark matter
  G. Tambalo, M. Zumalacárregui, L. Dai, M. Cheung · Phys. Rev. D 108, 103529 (2023)
- Gravitational wave lensing beyond general relativity: birefringence, echoes and shadows
  J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. D 102, 124048 (2020) · Editors' Suggestion
- Gravity in the era of equality: towards solutions to the Hubble problem without fine-tuned initial conditions
  M. Zumalacárregui · Phys. Rev. D 102, 023523 (2020)
- Limits on stellar-mass compact objects as dark matter from gravitational lensing of type Ia supernovae
  M. Zumalacárregui, U. Seljak · Phys. Rev. Lett. 121, 141101 (2018) · Editors' Suggestion
- Dark energy after GW170817: dead ends and the road ahead
  J. M. Ezquiaga, M. Zumalacárregui · Phys. Rev. Lett. 119, 251304 (2017) · Editors' Suggestion
- hi_class: Horndeski in the Cosmic Linear Anisotropy Solving System
  M. Zumalacárregui, E. Bellini, I. Sawicki, J. Lesgourgues, P. G. Ferreira · JCAP 08, 019 (2017)
- Transforming gravity: from derivative couplings to matter to second-order scalar–tensor theories beyond the Horndeski Lagrangian
  M. Zumalacárregui, J. García-Bellido · Phys. Rev. D 89, 064046 (2014)